

Where the 10 ms go at 100 Hz

One v2 policy tick on the robot's Raspberry Pi 3B, web interface running: 8.3 ms of work, 1.7 ms idle

3.47 ms

35 %

Policy inference

both neural nets 1.79 ms
+ gait generator + observation

2.23 ms

22 %

Web interface overhead

GIL contention: Flask,
IMU thread, recorder

1.90 ms

19 %

Safety governor

joint limits + winding
temperature observer

0.70 ms · 7 %

CAN I/O

once per tick:
drain + 6 frames

1.70 ms

17 %

Idle

Policy tick measured on the robot: 3.47 ms in its own process, 5.3–6.1 ms inside the web-UI daemon (2026-09-11); web overhead = the difference at the midpoint (1.8–2.6 ms).

Safety governor 1.9 ms and CAN send 0.4 ms per loop tick measured (2026-09-01); CAN receive + log 0.3 ms per loop tick inferred from the daemon's 3.0 ms non-controller budget.